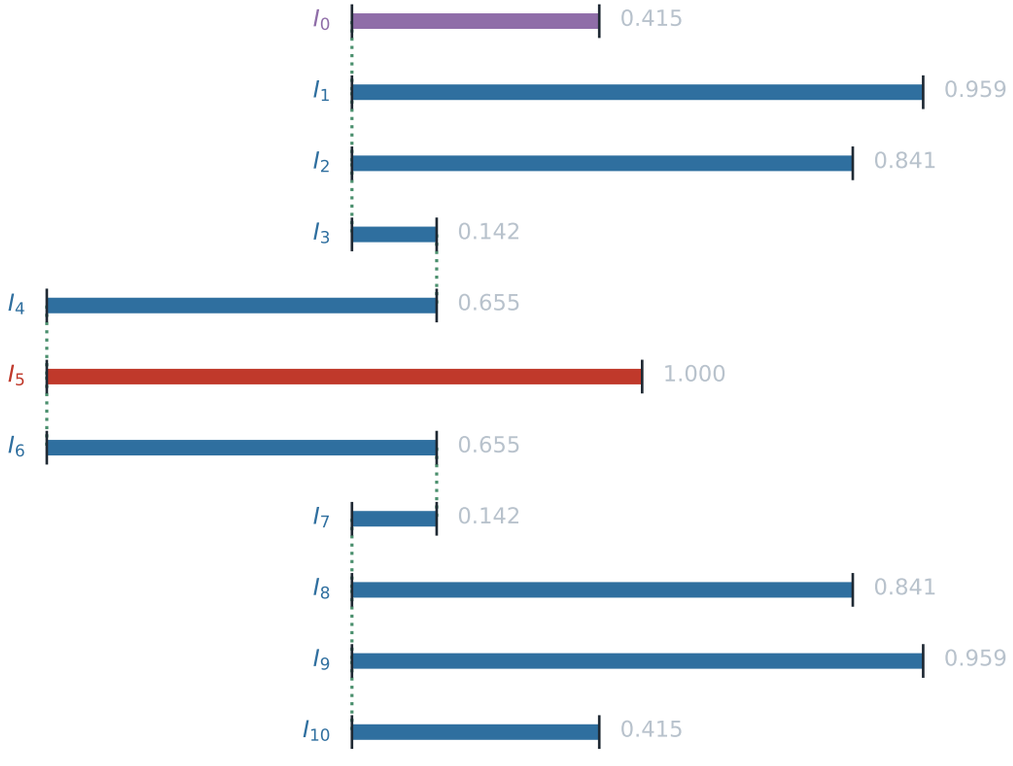
(a) DATA — the $Q$ interfaces in the global coordinate,
$I_t = [u_t,\ u_t + \ell_t]$. The offsets CLOSE around the cycle (Thm. 4.1).
Without that this picture would not exist.

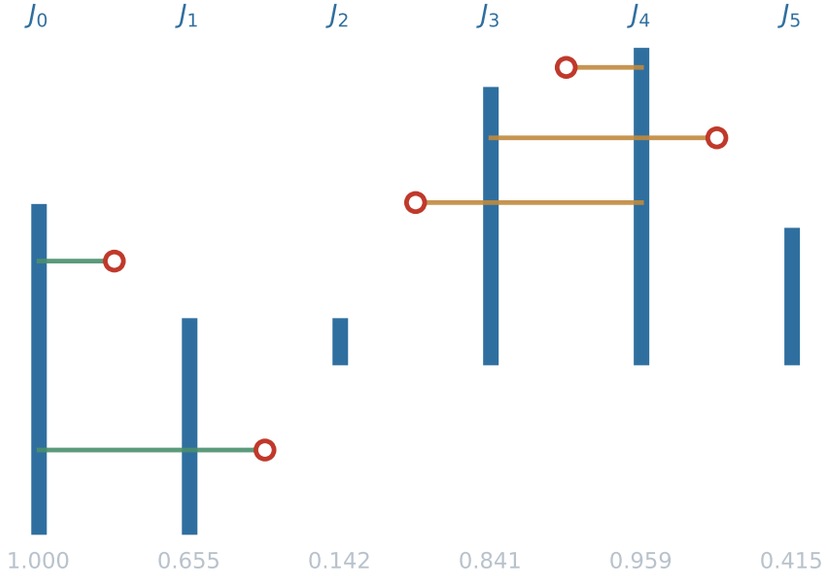



green dots: the shared endpoint of consecutive interfaces
(their intersection is the THROUGH band; the protruding
tail of the larger is where the FOLD lives)

the sequence is a PALINDROME about $t = m = 5$
and the kite-0 gap — that is $\iota$

(b) DATA — the $\iota$ quotient (Cor. 4.2): the path $J_0, \ldots, J_m$, $J_i := I_{m-i} = I_{m+i}$,
drawn in the same coordinate as (a). 3/11, PAIRED.

$|J_i| = |\cos(iP\pi/Q)|$ (PAIRED), $|\sin(iP\pi/Q)|$ (LONE)




one interior pole per EDGE of the path (5 of them), each at its fold's FIXED POINT
horizontal bars = the first run, at CONSTANT $y$: green = reaches an end of the path (2/5 here), gold = folds first

$p = 0 \implies$ every interior pole of the path escapes to an END of it (Corollary 4.2)
— a statement about $(Q+1)/2$ nested intervals; the converse is open in general.
There is no static criterion: escaping on the FIRST run is common and not universal.